

Connected Loads [kW] vs. Actual Phase Balance Analysis



[UC10-01] - Awesense Energy Transition Platform Use Case

The Utility Problem

The GIS data quality is a top-priority issue for many electrical distribution utility professionals. Among the GIS data quality issues, meter-to-phase assignment (or meter-to-phase identification) is one of the most prominent. Errors in meter-to-phase assignment mean that the utility system record contains wrong information about what phase the energy consumer is connected to. Errors in meter-to-phase assignment adversely influence grid operations and outage management or can lead to misjudgement in terms of Asset Management, for example, connection request handling.

To solve and correct meter-to-phase assignment, the utilities use multiple different approaches. These approaches vary in multiple aspects such as effort needed, scale or resolution. One of the broadly used approaches in meter-to-phase assignment correction is to assess the level of wrongly assigned meters on a feeder level. This method involves a comparison of feeder-connected kW per phase vs. phase balance load at the top of the feeder.

Analysis based on this approach allows utilities to target areas where effort for detection, verification and correction have the highest impact. Such analyses allow utilities to allocate resources where they will achieve the highest added value.

The Data Solutions and Methods

To describe the methodology of this analysis in general terms, it is necessary to mention the required data sets which are used for this type of analysis. These datasets are:

1. Information about what loads (consumers, producers or prosumers) are connected to the feeder and their phase information.
2. Consumption/generation or net load time-series data from all connected loads (consumers, producers or prosumers).
3. Feeder level phase balance (breaker, recloser or another SCADA device) from the top of the feeder.

After the collection of all required data and after the assurance that data are correct and synchronized, the process of getting the results of such analysis involves several steps. These steps are summing all load usage time-series per phase on the feeder and compare this information with the top of the feeder phase balance. Feeders experiencing the highest discrepancy should then be targeted for further actions involving finer-granularity meter-to-phase error detection, verification and correction.

With the Awesense Energy Transition Platform conducting such analysis is significantly simplified. Awesense Energy Data Model's (EDM) data model and data access functions are the main aspects making such analysis easy to deliver.

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1. The core dataset for Awesense Energy Transition Platform, the GIS data, is modelled in a versatile data model which recognizes notions like logical connectivity or phase assignation. In addition, the EDM's unique GIS querying functions allow analysts to simply obtain all connected assets to any point (all respecting connectivity and switchable element statuses). In this case, the function would retrieve all connected meters on the feeder alongside the information about the meter's phase assignation.
2. The consumption/generation or net load time series can be easily obtained as well due to EDM's built-in linkage of time series to the underlying GIS data. For analysts, a hefty function is available at their disposal. The result is a time series which can be turned to net load (if necessary) and then summed based on phase assignation obtained from the previous step.
3. To get feeder level phase balance, the process is quite similar to the previous one. This is due to the fact that in EDM, all time series are stored and modelled in a similar fashion. If the analyst knows IDs of the time series generating elements (usually breaker, recloser or another SCADA device) they can be served to the time series retrieval function, or if these IDs are not known, the analyst can explore EDM's time series data sources and obtain the necessary IDs from there.

Finishing this analysis is just a question of building the algorithms in a way that will produce summed kW per phase or directly determining the discrepancies for an analyzed feeder or feeders.

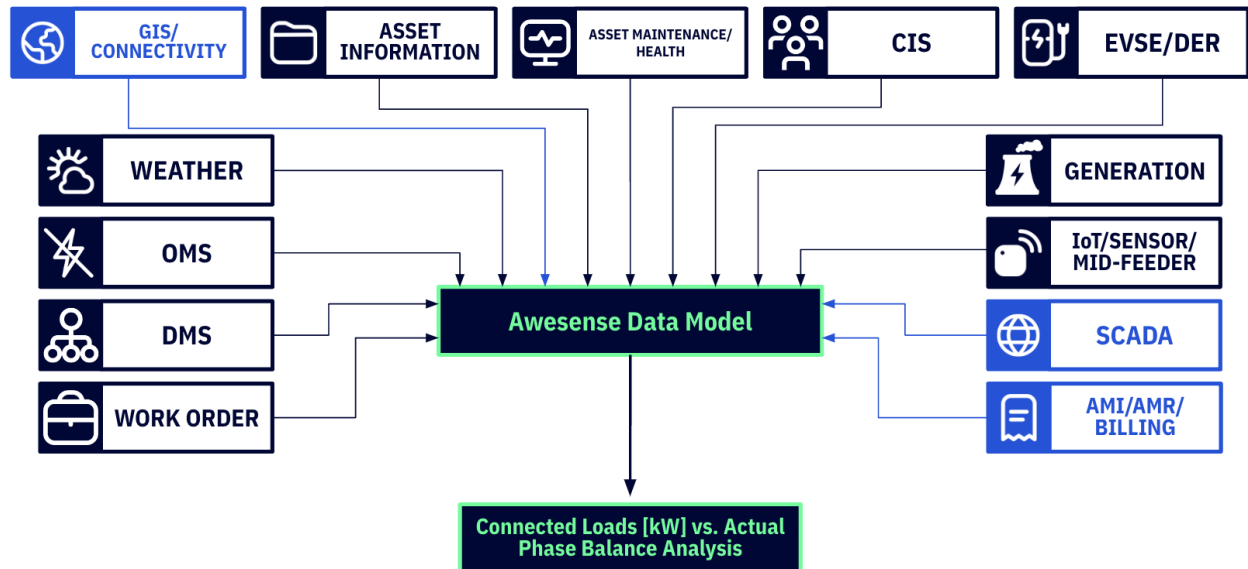
Required Data Sets

The required data for this use case are visualized in the following diagram. Data sources highlighted in [royal blue](#) are required for this use case.

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Available Use Case Implementations:

Users of the Awesense Energy Transition Platform can access this use case by the following means:

1. GitHub link to the Jupyter notebook can be provided upon request.